OLIVER ERNSTER BSc (Hons)

Principal Python Engineer & Software Architect: APIs, Integration & Platform Modernisation

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PROFILE

Software engineer and architect with 28+ years at the seams of systems: designing APIs and protocols, integrating platforms and joining legacy estates to modern ones without breaking the consumers that depend on them. The most recent 13+ years centre on Python backend, API and platform engineering across defence, fintech, telecoms, cloud and startups. I work hands-on while owning backend architecture and deployment, making structural decisions early so the right ones compound; a theme developed formally in my four-book *Decision Architecture* series on how decision flow and authority boundaries shape engineering outcomes.

I designed and specified the open protocol MMSP and spent ten years in telecoms conformance at Anite working on the 3GPP/ETSI GSM mobile conformance specifications and their TTCN-2 test suites, later developing UMTS protocol software, identifying defects in the published standards themselves and advising ETSI and other standards bodies, test houses, handset manufacturers and network operators on the corrections. I have repeatedly been brought into unfamiliar codebases to modernise, integrate and hand over cleanly.

I work fluently with advanced AI tooling (RAG, agentic workflows, multiple LLMs) through a professionally designed harness, with design decisions and production behaviour firmly my own. After a series of short, intensive delivery engagements across consultancies, startups and enterprise clients, I seek a permanent Staff/Principal engineering, integration/solutions architecture or modernisation-lead role with long-term ownership of architecture and technical direction.

Security: open to SC-cleared roles; eligible for UK SC clearance (not currently held), with prior defence-sector delivery at DERA, Northrop Grumman, Thales and Hewlett Packard.

CORE SKILLS

- **Leadership:** technical direction, stakeholder alignment and delivery planning; multidisciplinary teams of up to 17 at HP; Team Lead and Project Manager across projects at Anite; offered management of its Japanese office; the bridge between engineering, product and customer.
- **Standards, conformance & compliance:** ten years finding and correcting defects in published 3GPP/ETSI GSM conformance specifications and TTCN-2 suites, advising ETSI, other standards bodies, test houses, handset manufacturers and operators; author of the MMSP protocol specification with a conformance suite covering every normative statement; and PCI DSS compliance.
- **APIs, interfaces & integration (28+ yrs):** designing, consuming and integrating software interfaces across Python, C/C++, JavaScript/TypeScript and other stacks; API replacement and platform-to-platform integration with continuity preserved for downstream consumers.
- **Architecture & modernisation:** legacy migration (Python 2 → 3, MuleSoft → FastAPI), dependency rationalisation, re-platforming, idempotent deployment and production hardening; owning system structure from first principles through to handover.
- **Python (most recent 13+ yrs):** APIs, backend services, automation and production tooling; strong object-oriented and functional design-pattern experience.
- **Cloud & platform:** AWS (Lambda, CDK, Zappa), Terraform, Docker, CI/CD, deployment automation; serverless architecture defined from first principles.
- **Frameworks & data:** FastAPI, Flask and Django/DRF; SQL across PostgreSQL, MySQL, MariaDB, SQLite and MS SQL Server; NoSQL with Redis, MongoDB and Cassandra.
- **Full-stack & cross-platform:** end-to-end delivery from Python services through Vue/React/TypeScript front ends; development on Windows, macOS and Linux with packaged releases (Flatpak, signed .dmg, .exe).
- **Go (Golang) via AI-assisted workflows:** several working cross-platform applications delivered with Wails and React front ends on macOS, Linux and Windows 10/11; a public example is PigeonPost (pigeonpost.ink). Architecture, quality and production behaviour owned by me; I direct AI through Go effectively while making no claim to unaided Go fluency.
- **Foundations:** 15 yrs C++ and 4 yrs C across multithreaded, protocol-oriented and cross-network systems; retained depth, not my current focus.

PUBLISHED WORK & PORTFOLIO

Author of five books (Amazon; crankthecode.com/books): the four-book *Decision Architecture* series on decision flow, authority boundaries and organisational design (also as a hardback *Complete Series Edition*), and *Evergence: A Framework for Human-AI Co-Evolution*. The series is made operational in two open-source desktop tools: *Decitect* (decitect.com) scores an organisation's structure from 0 to 100 and plans the moves that strengthen it; *Simquence* (simquence.com) simulates an architecture's latency at design time, reporting p50–p99 before code exists.

Public projects include *Meridian* (reference client for MMSP) and *ClearBudget* (day-by-day solvency model), plus *NarrateX* (local text-to-speech reader), *Stellody* (local music player and discovery) and *PigeonPost* (Go/Wails desktop), spanning desktop apps, protocols, web systems and tooling. All open source: 60+ public repositories at github.com/oernster, with ernster.dev as the guided entry point. Commercial licences are available for every application (ernster.dev/commercial-licensing.html).

CAREER HISTORY

Cisco, *Senior Python Developer (Consultancy)* **2025**

- Brought a legacy Python codebase up to a current, maintainable standard: migrated Python 2 → 3 and re-platformed services onto FastAPI while keeping production accountable and running.

- Improved release predictability via platform upgrades, dependency rationalisation and idempotent deployment; established a modern React/TypeScript interface onto the modernised backend.
- Accelerated modernisation and architectural iteration through agentic AI-assisted workflows under professional control.

Tech: *Python 2/3, FastAPI, React/TypeScript, CI/CD, AI-assisted workflows.*

Harrier, *Lead Python Developer (Startup)* **2024-2025**

- Owned architecture end-to-end under startup constraints: delivered a serverless Python/FastAPI backend for ingestion, processing and reporting with no standing application infrastructure.
- Defined the AWS estate in Terraform from first principles and owned reliability and system behaviour in production.

Tech: *Python, FastAPI, AWS (serverless/Lambda), Terraform, SQL.*

LTIMindtree, *Senior Data Analyst (Consultancy)* **2023**

- Replaced a MuleSoft API with a maintainable FastAPI Python 3 platform, reducing operational complexity and cost while preserving continuity for every downstream consumer through the switchover.
- Improved architecture, coding standards and documentation to leave a clean, operable handover.

Tech: *Python 3, FastAPI, AWS, SQL, git.*

EPAM, *Lead Developer (Consultancy)* **2022**

- Prototyped a cloud-based flood-insurance API in a regulated environment, de-risking the architecture across FastAPI, Lambda, PostgreSQL, SQLAlchemy and AWS CDK and giving stakeholders a concrete route for decision-making.

Tech: *Python, FastAPI, AWS Lambda/CDK, PostgreSQL, SQLAlchemy, TypeScript.*

MDRxTech, *Senior Developer (Consultancy)* **2022**

- Delivered Python/FastAPI backend work across Docker, serverless/Lambda and SQL-backed systems for music-technology services.

Tech: *Python, FastAPI, Docker, AWS Lambda, SQL.*

Thales, *Senior Software Engineer (Cyber Resilience & Defence Research, Consultancy)* **2021**

- Enabled cyber-resilience modelling and visualisation for research users through a multithreaded socket-based backend joined to a Flask front end.
- Built a targeting web tool with variable range handling communicating with an encrypted backend API.

Tech: *Python, Flask, React, sockets/multithreading, SQLite.*

Unilink, *Senior Developer* **2021**

- Delivered reliable Django/DRF software supporting secure family communication under operational and regulatory constraints.

Tech: *Python 3, Django, Django REST Framework, Docker, MySQL.*

Validus Risk Management, *Senior Developer (FinTech)* **2020-2021**

- Enhanced a regulated wealth-management and risk-analysis platform in line with analytical requirements and business objectives.

Tech: *Python 3, Django/DRF, AWS, React/TypeScript, Docker, SQL, Linux.*

Memset, *Senior Developer (Cloud Platform)* **2019-2020**

- Strengthened a core Python/Django cloud platform and contributed to platform-level architecture in a pure cloud-engineering team.

Tech: *Python 2/3, Django, Linux, SQL, git.*

EARLIER CAREER (1997-2019): SELECTED ROLES

Anite, *Telecoms Conformance - Integration Engineer to Developer, Team Lead & Project Manager* **1998-2008**

- Worked across GSM, GPRS and UMTS mobile-station conformance, inter-operability and development test systems, chiefly the Stand Alone Tester (SAT), the company's GSM Phase 2 conformance product and main revenue earner, which I demonstrated internationally. Identified defects in the published 3GPP/ETSI GSM specifications and their TTCN-2 suites, advising ETSI, other standards bodies, test houses, handset manufacturers and network operators; later developed UMTS protocol software.
- Promoted from Integration Engineer to Project Manager, then into development by choice, declining an offer to manage the Japanese office; later Team Lead and Project Manager from an engineering footing. Four years of C and much of my C++ foundation built here.
- Turned a two-word brief from the Deputy MD, "network licensing", into a shipped product as lead developer and project manager: customer checkout/usage UI, secured internal licence generator, deployable licence checker. Originated and shipped the entitlement model, adopted by the company: products defined as sets of numbered licence codes, composable by aliasing (a higher-tier product entitling lower-tier ones, each still separately sellable and licensable) or grouping (same-tier bundles).
- Previously owned the company-wide local licensing solution (dongles and licence files); wrote most of PTGUI (Programmer's Toolset GUI, MFC/CArchive), prototype for the replacement network-operator interoperability test tool for dual-RAT and MultiRAT.

Tech: *C, C++, MFC/COM/ATL, multithreading, out-of-process servers/dongles, TTCN-2, GSM/GPRS/UMTS conformance, ETSI/3GPP liaison.*

- **Fintech (2016-2018):** Bink, GlintPay and Duologi - Python/Django systems across loyalty, payments and lending startups.
- **Media & broadcast (2012-2019):** MPC Advertising, Forbidden and EditShare - Python/Django, C++, Qt and Objective-C.
- **Defence & research:** Hewlett Packard (2012-2014), technical lead / Scrum Master for a 17-person multidisciplinary team; Northrop Grumman (2009-2010) and DERA (1997-1998) - C/C++, multithreaded and protocol-oriented systems.
- **Telecoms:** Semafone (2010-2011) - secure payment-by-phone under PCI DSS; C/C++/Python with SIP/ISDN telephony.
- **Security, IoT & other:** Armour Communications (secure comms, 2015-2016), PhotonStar (IoT, 2014), Artfinder (2012), Electric Motor Works (EV, 2008-2009), Home Office NGO (2010).

EDUCATION

BSc (Hons) Physics, University of Manchester. Final-year project: Infra-red reconnaissance satellite design.

INTERESTS

- Self-directed postgraduate-level physics study; authoring books: five published, with more planned.
- 3D printing (8 years): hardware, electronics, firmware and software across ten printers; Foundation amateur radio (ham) licence; Grade 7 piano (ABRSM); flight simulation with full HOTAS; reader of science-fiction and popular science.